Getting Started with Scaleform Video

Scaleform 4.3 Scaleform Video

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: 3.01

: 2011 9 15

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Autodesk Scaleform [illegible]:

[illegible]	Getting Started with Scaleform Video (Scaleform Video [illegible] 1)
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[illegible]	info@scaleform.com
[illegible]	(301) 446-3200
Fax	(301) 446-3199

目次

はじめに

CRI™ Autodesk® Scaleform® Video™ Scaleform Adobe® Flash® Flash HUD

CRI Movie CRI Movie CRI Middleware 1700

Scaleform Video PC (Xbox 360® PLAYSTATION®3 (PS3™) Wii™) Scaleform Video

Scaleform Video Adobe Premiere® Adobe After Effects® Adobe Flash 5.1 Scaleform Video

Scaleform Video SDK :

Scaleform Video

Scaleform Video

Getting Started with Scaleform Video ( )

CRI Movie Scaleform Video Scaleform Video : CRI
CRI Movie™ for Scaleform

について

Scaleform Video

USM

Flash

Video

Scaleform Player *videodemo.swf*

2.1

Scaleform Video :

C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Tools\VideoEncoder

Scaleform Video :

C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Data\AS2\Video

3

2.2

Encoder

getting_started_with_video_tutorial.fla – Flash

getting_started_with_video_tutorial.swf – Flash

sample.avi –

sample_audio.wav –

sample_cue_points.txt –

sample_subtitles.txt –

2.3 Encoder

Scaleform Encoder

Medianoche.exe —

ScaleformVideoEncoder.exe — ( )

TMPGLib.dll —

VideoEncoderUtil.dll — ScaleformVideoEncoder.exe

VideoPlayer.swf — Scaleform Video Encoder [Preview ( )]

2.4

API … (HD)
(1080p)

1

… SDK .

2.5

CRI Movie™ … Scaleform® Video … Adobe Flash® … Scaleform Video … 3D … HUD … Video … Scaleform Video … Scaleform 4.1 … Scaleform Video … CRI Movie™ … Adobe Flash

2.6

(AVI WAV)

: Scaleform Video Xbox 360 PS3 PC Wii

Render to Texture: 3D

(32)

: Scaleform 4.3 Flash
ActionScript™

: ActionScript
Scaleform Video

UI

5.1ch

- : Scaleform Video

(
)

[illegible]

Scaleform Video Encoder [illegible] USM [illegible]

[illegible] Flash [illegible]

[illegible] ActionScript [illegible]

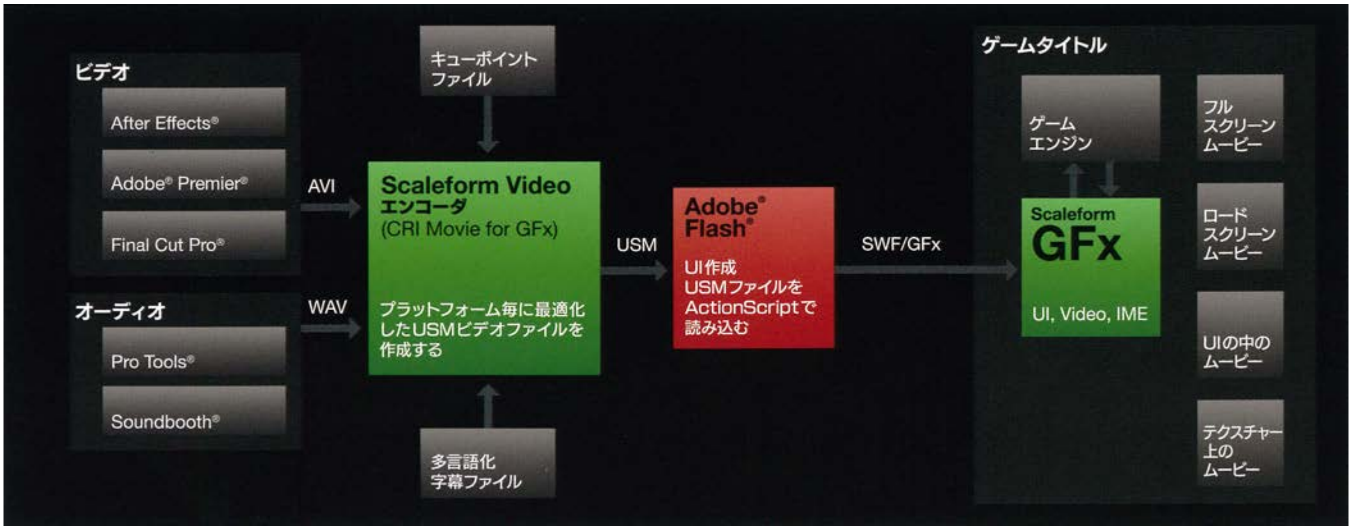
[illegible]

2.7

Scaleform Video USM Scaleform CRI Video USM Scaleform Media Player . Scaleform Player USM

C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Data\AS2\Video\ scaleform_logo.usm Player

2.8



ت 1: Scaleform Video ꒧ ꒱ ꒮ ꒧

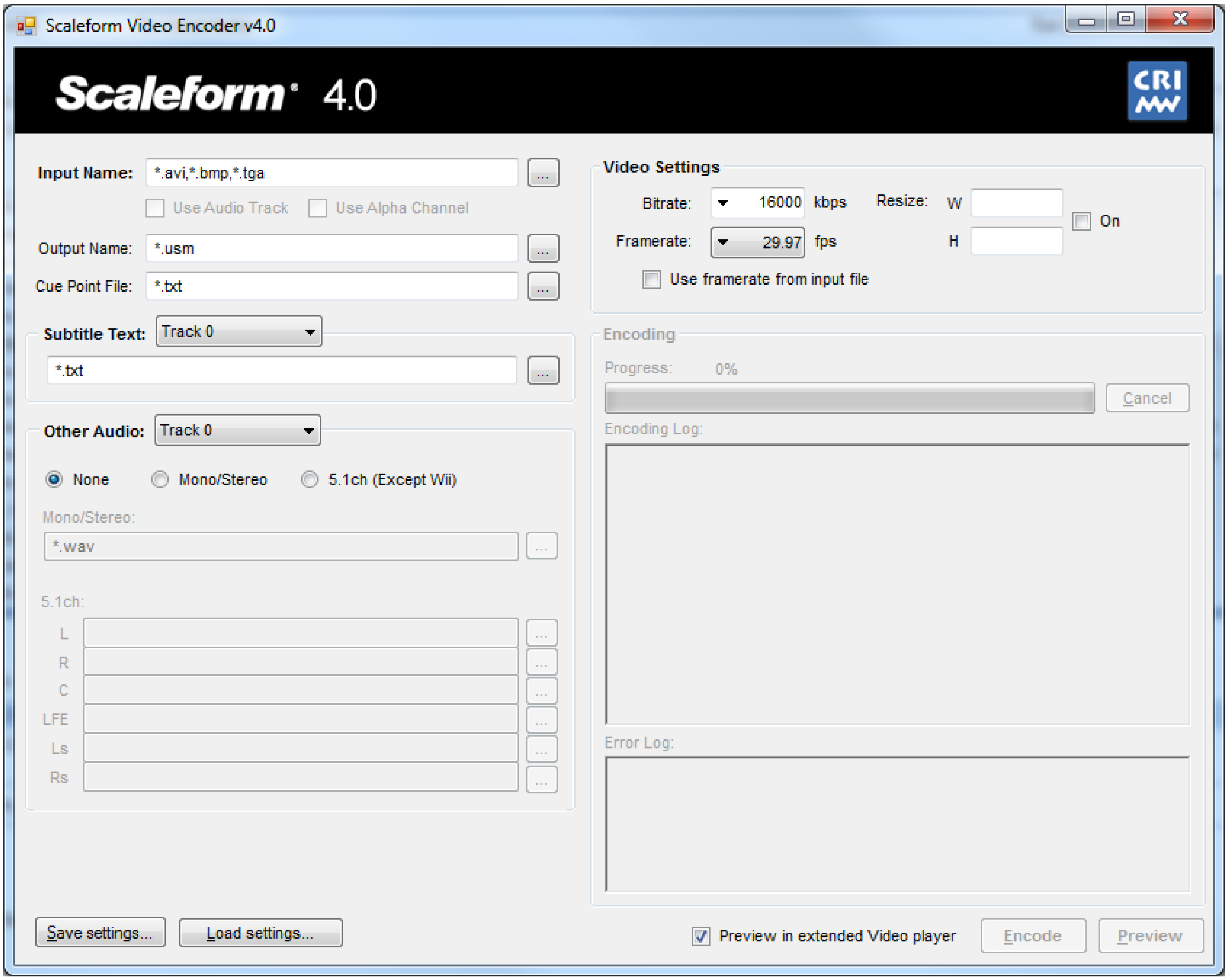
꒮ ꒣꒟ ꒒ ꒡ ꒱ δ ꜀ ꒧ ꒱ ꒮ ꒧ ꒛ . :

1. Scaleform Video API ˜ ꒮ ꒣ ꒱ ꒧ ꒮ ꒧ ꒛ .
2. Adobe Premiere ꒒ ꒡ ꒱ ꒱ ꒧ ꒮ 1 ꒡ ꒮ ꒡˜ AVI ꒮ ꒱ ꒡ ꒡
꒡ ꒱ ꒱ ꒮ ꒡ ꒡ .
3. Scaleform Video Encoder ˜ ٪... AVI ꒒ ꒡ ꒱˜ USM ꒮ ꒱ ꒡ ꒡ ꒡ ٢ .
4. USM ꒱ ꒧ ꒮ 1 ꒒ ꒡ ꒱˜ ꒸ ꒻ Adobe Flash SWF ꒮ ı 1˜ δ .
5. ꒒ ꒡ ꒱ ڳو ꒒ 1 ꒡ ꒧ ꒡ ˜ . ActionScript ˜ ꒛ .
6. ꒡ ꒮ ꒡ ꒡ ꒒ 1 SWF ꒮ ı 1˜ ꒮ ꒣ ꒧ ꒡ ꒡ .
7. ꒮ ꒣ ꒱ ꒧ ꒮ ꒧ ٪... ἰ ت - ꒛ ˜ ٪... ꒮ ꒣ SWF ꒮ ı 1˜ ֆ ꒧ .

* : USM ꒒ ꒡ ꒱˜ Flash ꒮ ı 1 ꒻ ActionScript ˜ ٪... . ˜ ꒡ .
꒒ ꒡ ꒱˜ ꒡ ꒡ ꒡ ꒡ ꒡ ꒮ ꒣ ֆ ꒧ . ꒒ ꒡ ꒱˜ ꒮ 1 1 ꒡ ꒮
꒛ ꒛ ᲿᲦ ꒛ ꒧ ꒮ ı 1 U ֆ ꒱ ꒧ ꒮ 1 . Flash ꒮ ı 1 ج ꒛
꒒ ꒡ ꒱ o ج . .

クイック ガイド のビデオをエンコードする

Scaleform 4.3 ٪… ĩ . CRI USM . . ˜
CRI ă . ˓
. ă . . ˜ Scaleform Scaleform Video
Encoder ˜ ɥ ڗ Encoder ˜ ٪… USM ˜ 1 .
ط وگ ک



ت 2: Scaleform Video Encoder

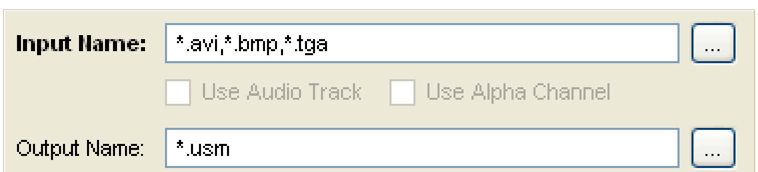
3.1

USM 2 . (2) USM

Windows™ [ ] Scaleform Video Encoder Scaleform Video Encoder . :

Windows Vista: [ ]->[ ]->[Scaleform]->[GFx SDK 4.3]->[Video]
Windows XP: [ ]->[ ]->[Scaleform]->[GFx SDK 4.3]->[Video]

1. [ ] 2 . AVI
   a. AVI BMP . TGA .
   b. BMP TGA



3: . [Input Name] .

2. . [Use Audio Track ( )] AVI . BMP TGA .
3. USM . . [Use Alpha Channel ( )] . . .
4. [Output Name (U)] . . [ ] USM
5. [Encode ( )]

3.2

1. Encoder

2. *C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Data\Tools\VideoEncoder\sample.avi*

3. [Encode ( )]

4. [Preview ( )] USM

3.3

[illegible] SWF [illegible]

[illegible] USM [illegible] [Cue Point File ([illegible])] [illegible]

[illegible] 4: [illegible]

[illegible] 0 [illegible] ([illegible] 1000 [illegible]) [illegible] 0 [illegible] 5000 [illegible] 5 (5000 / 1000 = 5) [illegible]

[illegible] 2 [illegible] DVD [illegible] 0 [illegible] 1 [illegible] Flash [illegible] ActionScript [illegible]

[illegible]

```
time, cue point type (0 or 1), cue point name, parameter1, parameter2, …
paramter10
```

[illegible]:

```
1000, 0, cue_point_1, my_param=5, my_other_param=10
```

[illegible] CuePoint.txt [illegible]

```
1000
0,0,start
1000,0,cue1
2000,1,cue2,name1_0=value1_0,name1_1=value1_1
3000,0,cue3
4000,1,cue4,name4_0=value4_0
```

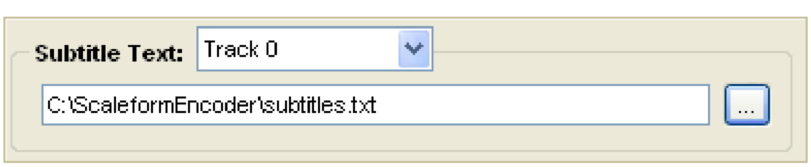
```
5000,0,cue5
6000,1,cue6,name6_0=value6_0,name6_1=value6_1
7000,0,cue7,name7_0=value7_0,name7_1=value7_1,name7_2=value7_2
8000,1,cue8,name8_0=value_0
9000,0,cue9
10000,1,cue10,name10_0=value10_0,name10_1=value10_1
11000,0,end
```

キュー ポイントを持つサンプル ビデオを再エンコードする

1. Scaleform Video Encoder
2. *C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Tools\VideoEncoder\sample.avi*
3. [Cue Point File ( )] [ ] *sample_cue_points.txt* *sample.avi*
4. [Encode ( )]
5. [Preview ( )] USM
6. Video

3.4

1. [Subtitle Text ( )] 32 (Track 0-31)
2. [ ]



5:

1000 ID 3 ID 1 Flash

ActionScript

start time, end time, messageID

サンプルの ファイルの内容

```
1000
2000, 5000, This is the first subtitle.
8000, 11000, subtitleId2
```

字幕を持つサンプル ビデオを再エンコードする

1. Scaleform Video Encoder
2. *C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Tools\VideoEncoder\sample.avi*
3. [Cue Point File ( )] *sample_cue_points.txt* *sample.avi*
4. [Subtitle Text ( )] *sample_subtitles.txt* *sample.avi*
5. [Encode ( )]
6. Scaleform Scaleform Media Player *getting_started_with_video_tutorial.swf* &
7. Video 2-5 8-11

3.5

Encoder 5.1

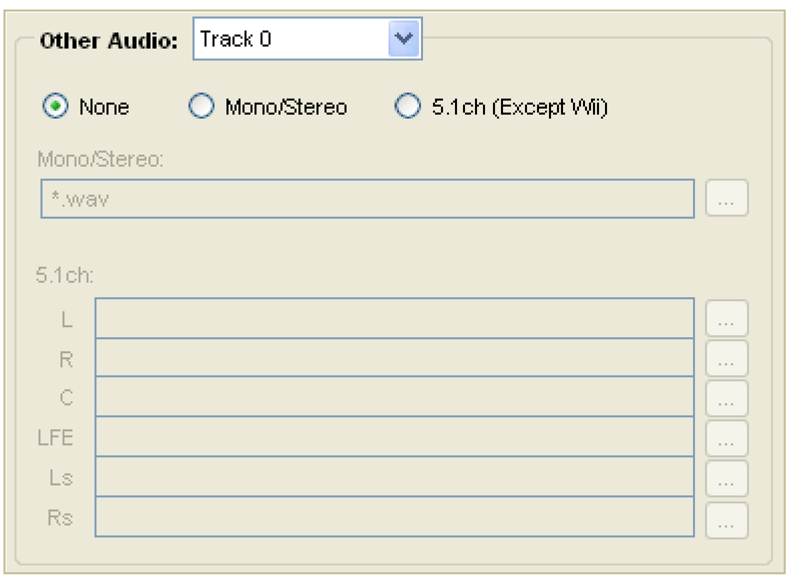
1. Scaleform Video Encoder [Other Audio ( )]

2. [None ( )] [Mono/Stereo (/)] [5.1ch (Except Wii) (5.1ch (Wii))]

   : 5.1 Wii

3. [Mono/Stereo (/)] [Mono/Stereo (/)] [ ] WAV

4. [Use Audio Track ( )]



6:

5. 5.1ch L R C LFE Ls Rs [ ]

5.1 :

L –

R –

C – ( )

LFE –

Ls –

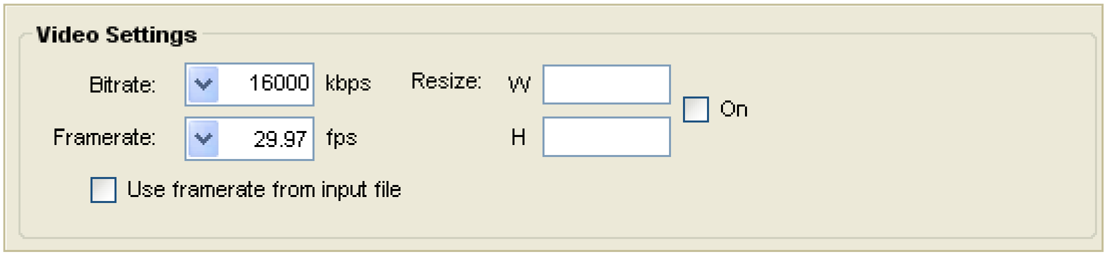
Rs –

つのオーディオトラックを持つサンプル ビデオを再エンコードする

1. Scaleform Video Encoder
2. *C:\Program Files\Scaleform\GFx SDK 4.3\Bin\Tools\VideoEncoder\sample.avi*
3. [Cue Point File ( )] [ ] *sample_cue_points.txt* *sample.avi* .
4. [Subtitle Text ( )] . [ ] *sample_subtitles.txt* *sample.avi* .
5. [Other Audio ( )] [Track 1]
6. [Mono/Stereo (/)]
7. [Mono/Stereo (/)] . [ ] *sampe_audio.wav* *sample.avi* .
8. [Encode ( )] : . . . . .
9. [Preview ( )] . . USM .
10. Video [Subtitle ( )] . [Audio Channel ( )]

3.6

[illegible] 1 [illegible] [Bitrate ([illegible])] [illegible]



[illegible] 7: [illegible]

1. [Bitrate ([illegible])] [illegible]

[illegible]

[illegible] (kbps)					
[illegible]		PS3	Xbox360	Wii	PC
1080p	[illegible]	40000	40000	[illegible]	40000
1080p	[illegible]	30000	30000	[illegible]	30000
1080p	[illegible]	20000	20000	n/a	20000
720p	[illegible]	36000	36000	n/a	36000
720p	[illegible]	26000	26000	n/a	26000
720p	[illegible]	16000	16000	n/a	16000
480p	[illegible]	16000	16000	8000	16000

480p	ى	12000	12000	6000	12000
480p	ﻝ	8000	8000	4000	8000

2. ( ) [Framerate ( )] ... 29.97 fps ى NTSC ... AVI Encoder ... [Use framerate from input file ( )] ... Encoder ... ð ... ð ... Շ NTSC ى ... 1 29.97 ... PC ... 1 60 ... 60 ...

3. ( ) [Resize ( )] [W (I)] [H (ﻝ)] ... I [W] ﻝ [H] ... [On] ...

4. [Encode (1)] ... [Preview ( )] ... ₯

   ... USM ... ð ... گ .

5. ... [Preview ( )] ... ₯

   [Encode (1)] ... [Preview in extended Video player (ث Video Player .)] ... ₯ ...

クイック ガイド ビデオを に追加する

: Flash . USM .
ActionScript SWF
.
.
Flash NetStream

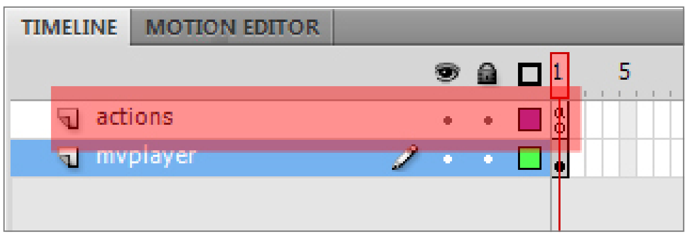
. Scaleform Encoder USM
1 : Scaleform .
. USM SWF SWF
Scaleform Player

4.1 Flash

Scaleform Video Encoder ... AS 2.0 Flash ... USM ... SWF ...

1. ... mvplayer ...

2. ... actions ...



8: ... actions ... mvplayer ...

3. ... *actions* ... 1 ... F9 ... :

```
_global.gfxExtensions = true;
_focusrect = false;
onLoad = function()
{
        mvplayer.playVideo("sample.usm");
}
```

... SWF ... USM ...

: *“C:/pathtovideo/sample.usm”*

4. ... *mvplayer* ...

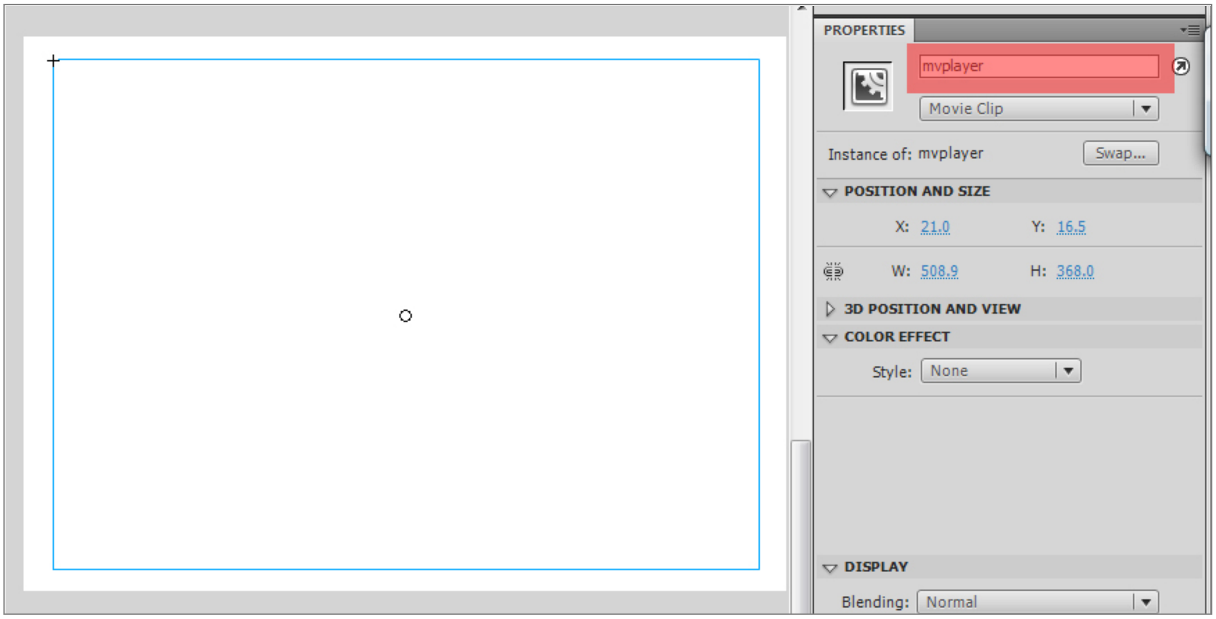
5. ...

   a. ... 1 ...

   b. ...

   c. ...

d. U mvplayer [OK]



9: mvplayer

6. mvplayer

7. [ ] U mvplayer

8. mvplayer

9. mvplayer actions

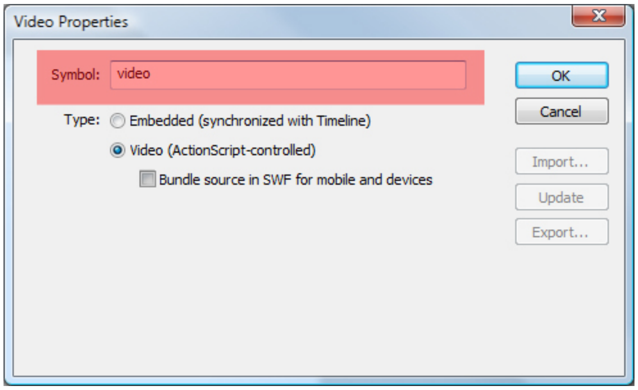
10. actions 1 [ ]
:

```
nc = new NetConnection();
nc.connect(null);
ns = new NetStream(nc);
video.attachVideo(ns);
this.attachAudio(ns);
var audio_sound:Sound = new Sound(this);
function playVideo(videoToPlay:String):Void
{
        ns.play(videoToPlay);
}
```

: . 1 mvplayer
actions 1 . .

11. [ ] [ ]

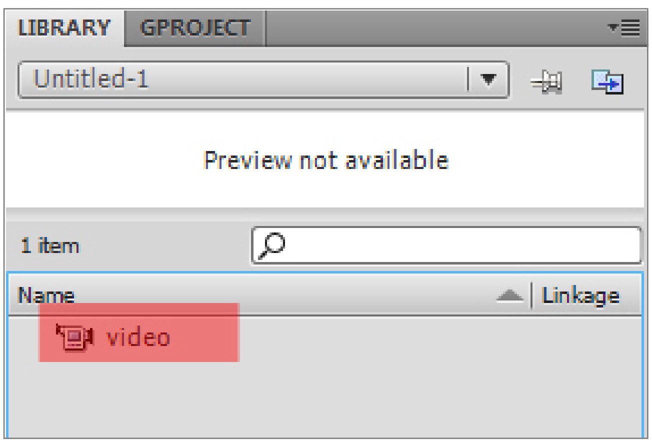
12. ᐱ ر ᑋ ᑎ ᑌ ¢video U ᐱ˜ ط [OK] ˜ ᑌ ᑎ ᑎ ᑌ



ت 10: ᑋ ᑎ ᑌ ᑌ ᑌ ᑌ˜ ᑌ ᑎ˜ ŏ .

13. [ᑋ˜ ᑎ ڿ] ᑌ ᑌ ᑌ˜ ٪... ᐱ ر ᑋ˜ ᑎ ˜ ŏ ᑋ˜ ᑎ ṡ ᑋ˜ ᑎ 3Ṣ
U ᐱ .

14. ṡ ᑋ˜ ᑎ 3Ṣ ᑌ ᑋ ᑎ 1 ᐱ ر ᑋ ᑎ ᑌ ᑌ ᑌ ᑌ˜ ᑌ ᑎ˜ [ᑎ˜ ᑌ ᑎ ᑎ] ᑌ ᑎ ᑋ ᑌ
ᑎ ᑌ 1 ᑎ ᑌ ᑌ ٕ ★ ح غ˜ ٢ ᑌ˜ ᑌ˜ ٢



ت 11: [ᑎ˜ ᑌ ᑎ ᑎ] ᑌ ᑎ ᑋ ᑋ ᑎ ᑌ ᑌ ᑌ ᑌ˜ ᑌ ᑎ

15. ᑌ ᑎ ᑌ ᐱ ᑋ ᑎ ᑌ ᑌ ᑌ ᑌ˜ ᑌ ᑎ˜ ˜ [ᑌ ᑋ ᑌ ᑎ˜] ᑌ ᑌ ˜ ᑋ ᑌ ᑌ ᑋ ᑌ U
˜ ¢video ᐱ

16. ᑎ ᑋ ˜ ᾰ ۶ ᑌ ᑌ ᑎ ᑌ ᑌ ᑋ

4.2

SWF . Scaleform Player .

1. Flash [Scaleform Launcher] SWF

1. Scaleform Player
2. SWF Scaleform Player &
   a. SWF Scaleform Player

クイック ガイド　でビデオを操作する

USM (
ActionScript Flash
: Flash
Flash

Flash SWF Scaleform
ActionScript

5.1 ActionScript

USM [illegible] Flash [illegible] Flash [illegible]

[illegible] USM [illegible] Flash SWF [illegible] sample.usm [illegible] Flash [illegible]

[illegible] ActionScript [illegible] mvplayer [illegible] actions [illegible] 1 [illegible]

1. *mvplayer* [illegible]
2. *mvplayer* [illegible] *actions* [illegible]
3. [illegible] traceCuePoint

```
ns.onCuePoint = traceCuePoint;
```

4. traceCuePoint [illegible] metaPropPJParams [illegible] (currentCuePoint.name) [illegible] (currentCuePoint.time) [illegible] (currentCuePoint.type) [illegible] (ns.time [illegible]) [illegible] :

```
function traceCuePoint(currentCuePoint:Object):Void
{
	var metaPropPJParams:Object = currentCuePoint.parameters;
	trace("\t\t name: " + currentCuePoint.name);
	trace("\t\t time: " + currentCuePoint.time);
	trace("\t\t type: " + currentCuePoint.type);
	if (metaPropPJParams != undefined)
	{
		trace("\t\t parameters:");
		traceObject(metaPropPJParams, 4);
	}
}
```

5. [illegible] traceObject [illegible] SWF [illegible] (๒) ๒ [illegible]

```
function traceObject(obj:Object, indent:Number):Void
{
	var indentString:String = "";
	for (var j:Number = 0; j < indent; j++)
	{
		indentString += "\t";
	}
	for (var i:String in obj)
	{
		if (typeof(obj[i]) == "object")
		{
			trace(indentString + " " + i + ": [Object]");
			traceObject(obj[i], indent + 1);
		}
		else
		{
			trace(indentString + " " + i + ": " + obj[i]);
		}
	}
}
```

6. [Scaleform Launcher] [illegible] SWF [illegible] Scaleform Player [illegible] Flash [illegible] Scaleform Player [illegible] Scaleform Player [illegible]

5.2 ActionScript

Flash

USM Flash SWF sample.usm

Flash

1. *mvplayer*
2. *mvplayer* *actions*
3. subtitleChannelNumber 0

```
var subtitleChannelNumber = 0;
```

4. 0

```
ns.onMetaData = function(infoObject:Object)
{
	ns.subtitleTrack = 1;
	subtitleChannelNumber = infoObject.subtitleTracksNumber;
	trace("Subtitle Channels: " +  subtitleChannelNumber);
}
```

5. *mvplayer* subtitle

6. msg USM subtitleId2 2 ActionScript msg 5 *subtitle*

```
ns.onSubtitle = function(msg:String)
{
	trace("Subtitle: " + msg);
	subtitle.text = msg;
}
```

7. [Scaleform Launcher] SWF Scaleform Player Flash Scaleform Player Flash

5.3 ActionScript

USM ... SWF ...

... USM ... Flash SWF ... Flash ...

mvplayer ... *actions* ...

1. ... *mvplayer* ...
2. *mvplayer* ... *actions* ...
3. 2 ... AudioTracks ... 2 ... CurAudioTrack ... 0 ...

```
var AudioTracks:Array;
var CurAudioTrack = 0;
```

4. ... audioTracks ... AudioTracks ...

   ... 'ns.onMetaData' ... 'AudioTracks = infoObject.audioTracks;' ... 'trace("Subtitle Channels: " + subtitleChannelNumber) ...

```
ns.onMetaData = function(infoObject:Object)
{
	AudioTracks = infoObject.audioTracks;
}
```

5. ... *mvplayer* ... prev_audiotrack ...

6. *mvplayer* *actions* [illegible] prev_audiotrack [illegible]

```
prev_audiotrack.onRelease = function()
{
      if (AudioTracks  == undefined || AudioTracks.length < 2)
      {
             return;
      }
      if (CurAudioTrack == 0)
      {
            CurAudioTrack = AudioTracks.length - 1;
      }
      else
      {
             CurAudioTrack--;
      }
      ns.audioTrack = AudioTracks[CurAudioTrack].trackIndex;
}
```

7. *mvplayer* [illegible] 5 [illegible] next_audiotrack [illegible]

8. *mvplayer* *actions* [illegible] next_audiotrack [illegible]

```
next_audiotrack.onRelease = function()
{
     if (AudioTracks  == undefined || AudioTracks.length < 2)
     {
            return;
     }
     CurAudioTrack++;
     if (CurAudioTrack >= AudioTracks.length)
     {
            CurAudioTrack = 0;
     }
     ns.audioTrack = AudioTracks[CurAudioTrack].trackIndex;
}
```

9. Flash [illegible] Scaleform Player [illegible] [NEXT] [PREV] [illegible] (USM [illegible] 1 [illegible])

のビデオ エクステンション

Scaleform ActionScript Flash
Scaleform
NetStream Adobe NetStream

6.1 NetStream

`currentFps:Number`

```
var currentFps:Number = ns.currentFps;
```

`time:Number`

```
var currentTime:Number = ns.time;
```

6.2 NetStream

`onCuePoint = function(infoObject:Object) {}`

USM

: Adobe Flash onCuePoint

`onMetaData = function(infoObject:Object) {}`

USM Player

: Adobe Flash onMetaData

audioTracks

USM

```
ns.onMetaData = function(infoObject:Object)
{
	audioTrackArray = infoObject.audioTracks;
}
```

subtitleTracksNumber

USM

```
ns.onMetaData = function(infoObject:Object)
{
      numSubtitleChannels = infoObject.subtitleTracksNumber;
      trace("Subtitle Channels: " +  numSubtitleChannels);
}
```

```
onStatus = function(infoObject:Object) {}
```

NetStream . .

: Adobe Flash onStatus

6.3 NetStream

```
close()
```

ns.time 0 .

```
: ns.close();
```

```
pause([flag:Boolean])
```

```
: ns.pause(true);
```

```
play(name:Object, start:Number, len:Number, reset:Object)
```

(USM)

```
: ns.play("myVido.usm");
```

```
seek(offset:Number)
```

```
: ns.seek(50);
```

```
setBufferTime(bufferTime:Number)
```

. GFx::Video::Video . . . . . 1 . .

```
ns.setBufferTime(2.5);
```

6.4 Scaleform

`subtitleTrack`

NetStream

```
ns.subtitleTrack = 1;
```

`subtitleTrack = Number`

NetStream

```
ns.subtitleTrack = 1;
```

`audioTrack = Number`

NetStream

```
ns.audioTrack = 2;
```

`subAudioTrack = Number`

SubAudio NetSream

```
ns.subAudioTrack = 3;
```

`voiceTrack = Number`

5.1 5.1 1 1

```
ns.voiceTrack = 1;
```

`setReloadThresholdTime(reloadTime:Number)`

setBufferTime 4 1 1 1 1 1 GFx::Video::Video 4 3 1 リロード スレッショルド GFx::Video::Video

```
ns.setReloadThresholdTime(1);
```

`setNumberOfFramePools(numPools:Number)`

GFx::Video::Video 1 1 1 CPU 2

```
ns.setNumberOfFramePools(3);
```

```
setOpenTimeout(timeout:Number)
```

USM 5 0

```
ns.setOpenTimeout(5);
```

```
currentFrame:Number
```

```
trace("Current FPS: " + ns.currentFps + " fps");
```

```
Loop = Boolean
```

loop true NetStream loop false

```
ns.loop = true;
```

```
clipRect = new rectangle(x:Number, y:Number, width:Number, height:Number)
```

```
ns.clipRect = new rectangle(100, 100,  240, 200);
```

```
onSubtitle = function(msg:String) {}
```

Player USM

```
ns.onSubtitle = function(msg:String) { trace(msg); }
```

ビデオの作成を理解する

1080p Xbox 360 PS3™

1080p CG

720p 1080p

Scaleform Video Encoder

Scaleform Video AVI Scaleform Video AVI

(DIVX)

Scaleform Video

5.1

Scaleform Video

1

7.1

Scaleform Video

2

技術的統合について

Scaleform Video

8.1 Scaleform

Scaleform GFx::Audio GFx::Loader SoundRenderer SWF SoundRenderer C++ Scaleform FMOD Sound::SoundRendererFMOD::CreateSoundRenderer FMOD::System

:

```
//

FMOD::System* pFMOD = NULL;
result = FMOD::System_Create(&pFMOD);
if (result != FMOD_OK)
      exit(1);
result = pFMOD->getVersion(&version);
if (result != FMOD_OK || version < FMOD_VERSION)
      exit(1);
result = pFMOD->init(64, FMOD_INIT_NORMAL, 0);
if (result != FMOD_OK)
      exit(1);
Ptr<SoundRenderer> psoundRenderer =
                        *SoundRendererFMOD::CreateSoundRenderer();
if (!psoundRenderer->Initialize(pFMOD))
      exit(1);

// GFx::Audio

Ptr<GFx::Audio> paudio = *new GF::xAudio(psoundRenderer);
loader.SetAudio (paudio State);
```

8.2 Scaleform

Scaleform … Video … GFx::Loader … VideoSoundSystem … GFx::Video::Video … Scaleform … SoundRenderer …

例

```
// … Video …
// …
Ptr<GFx::Video::Video> pvc = *new Video (Thread::NormalPriority);
// SoundRenderer …
// …
if (psoundRenderer)
{
        pvc->SetSoundSystem(psoundRenderer);
}
// …
// …
// : SetSoundSystem … Video …
 pvc->SetSoundSystem(Ptr<Video::VideoSoundSystem>(*new
                                    VideoSoundSystemDX8(0)));

loader.SetVideo(pvc);
```

Windows … Video::Video … VideoPC … 3 …

```
VideoPC(Thread::ThreadPriority
          decodingThreadsPriority = Thread::NormalPriority,
          int decodingThreadsNumber = MAX_VIDEO_DECODING_THREADS,
             UInt32* affinityMask = NULL);
```

例：

```
UInt32 affinities[] = {
                     DEFAULT_VIDEO_DECODING_AFFINITY_MASK,
                     DEFAULT_VIDEO_DECODING_AFFINITY_MASK,
                     DEFAULT_VIDEO_DECODING_AFFINITY_MASK };

Ptr<Video::Video> pvc = *new Video::VideoPC(Thread::NormalPriority,
```

```
MAX_VIDEO_DECODING_THREADS, affinities)
```

PS3 Xbox360 Scaleform Video

PS3 GFx::Video::VideoPS3 4 :

```
VideoPS3(CellSpurs* spurs, int spu_num,
              int ppu_num, Thread::ThreadPriority ppu_priority,
              int spu_num, UInt8* spurs_priorities = NULL );
```

spurs –	CellSpurs
spu_num –	SPU
ppu_num –	PPU
ppu_priority –	PPU
spurs_priorities –	SPU

例

```
UInt8 spurs_priorities[] = {1,1,1,1,0,0,0,0};
Ptr<Video::Video> pvc = *new VideoPS3(GetSpurs(), 0, 1000,
                                      4, spurs_priorities);
```

利用率の低減例：

PPU Scaleform Video
PPU 5 720p 50 60FPS

PPU 40 PPU
4 5
1

```
UInt8 spurs_priors[] = {0,0,1,1,1,1,0,0};
Ptr<Video::Video> pVideo = *new VideoPS3(GetSpurs(), 0,
                         Thread::BelowNormalPriority, 4, spurs_priors);
```

Xbox360 VideoXbox360

[illegible]

```
VideoXbox360(unsigned deCodingProcs = Xbox360_Proc1 | Xbox360_Proc3 |
                        Xbox360_Proc5, Thread::ThreadPriority
                        decodingThreadsPriority = Thread::NormalPriority,
                        Xbox360Proc readerProc = Xbox360_Proc2);
```

例

```
Ptr<Video::Video> pvc = *new VideoXbox360(Xbox360_Proc1 |
                                          Xbox360_Proc3|Xbox360_Proc5,
                                          Xbox360_Proc2);
```

8.3 [illegible] API

[illegible] Video::VideoBase::ReadCallback [illegible] VideoBase::IsIORequired VideoBase::EnableIO [illegible]

ReadCallback [illegible]

例：

```
// Simple read callback implementation for the video system
class VideoReadCallback : public VideoBase::ReadCallback
{
      public:
      VideoReadCallback() {}
      virtual ~VideoReadCallback() {}

      virtual void OnReadRequested()
      {
          // Video read started
          VideoReadStart = Timer::GetTicks();
          // ...
      }
      virtual void OnReadCompleted()
      {
          // Video read completed
          GameReadStart = Timer::GetTicks();
          // ...
```

```
        }

        UInt64  VideoReadStart;
        UInt64  GameReadStart;
    };

    // Init video system
    Ptr<Video::Video> pvc = *new Video::VideoPC(
            Thread::NormalPriority, MAX_VIDEO_DECODING_THREADS, affnts);

    // Create and set video read callback
    Ptr<VideoReadCallback> pvideoReadCallback = *new VideoReadCallback;
    pvc->SetReadCallback(pvideoReadCallback);
```

IsIORequired … true … IO …
EnableIO …

EnableIO …

例：

```
    if(pvc->IsIORequired())
    {
        pDataLoader->Enable(false);
        pvc->EnableIO(true);
        fprintf(stderr, "Video is loading\n");
    }

    // ...

    if(!pvc->IsIORequired())
    {
        pvc->EnableIO(false);
        pDataLoader->Enable(true);
        fprintf(stderr, "Game data is loading\n");
    }
```

… Apps\Samples\Video … 1 … 2 … VideoBGLoadFlash VideoBGLoadData …

- VideoBGLoadFlash - … Flash … 1 …
- VideoBGLoadData – … 1 …

8.4 Scaleform VideoSoundSystem

GFx::Video::VideoSoundSystem Video::Video … Video::SetSoundSystem() …

Video …

- Video::VideoSoundSystemDX8 – Windows DirectSound
- Video::VideoSoundSystemXA2 – Windows Xbox 360 XAudio2
- Video::VideoSoundSystemPS3 – PS3 MultiStream
- Video::VideoSoundSystemWii – Wii …
- Video::VideoSoundSystemFMOD – FMOD …
- Video::VideoSoundSystemWwise – Wwise …

Scaleform … Flash … Scaleform … Sound::SoundRenderer 8.1 … Video::VideoSoundSystem … SoundRenderer … VideoSoundSystem … VideoSound … SoundRenderer … SoundRenderer … Video … SoundRenderer … 2 …

```
#include "Video/Video_VideoSoundSystemXA2.h"
pvc->SetSoundSystem(Ptr<Video::VideoSoundSystem>(*new VideoSoundSystemXA2(0, 0)));
```

… GFx::Video::VideoSoundSystem GFx::Video::VideoSound … VideoSoundSystem … VideoSoundSystem … VideoSound … Create … Scaleform … VideoSound … Start Stop … Scaleform … VideoSound::PCMStream … VideoSoundSystem …

Audio Input … VideoSoundSystemWwise … v2009.2.1 build 3271 … Wwise SDK … Audiokinetic … Visual Studio … SDK\samples\Plugins\AkAudioInput … Audiokinetic … Wwise … GFx::Video::Video … Wwise SDK …

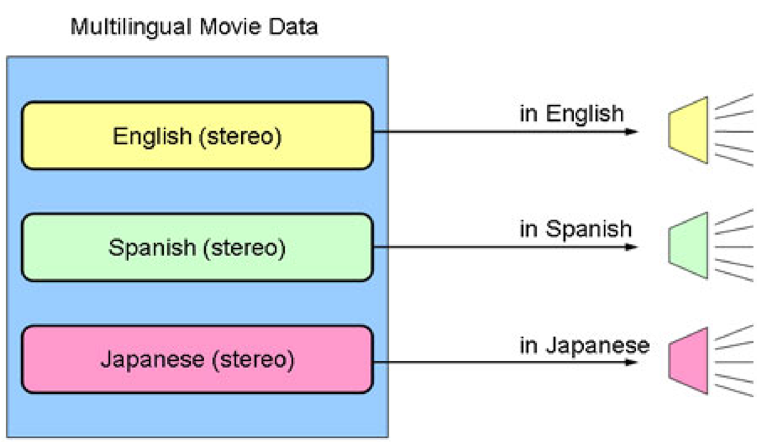
```
#include "Video\Video_VideoSoundSystemWwise.h"
Ptr<Video::VideoSoundSystem> wwise =
                        Ptr<Video::VideoSoundSystem>(*new VideoSoundSystemWwise());
pvc->SetSoundSystem(wwise);
wwise->Update();
```

Apps\Samples\FxPlayer … *Apps\Samples\Common* … Scaleform Player … FxPlayerAppBase.cpp … FxSoundFMOD.cpp … FxSoundWwise.cpp … Video_VideoSoundSystem*.cpp … *Src\Video* …

8.5

GFx::Video::Video … ActionScript … NetStream … *audioTrack* …

```
ns.audioTrack = 2;
```



： 多言語ビデオ

5.1

GFx::Video::Video 2

1.

2. SubAudio

センター チャネルの置き換え

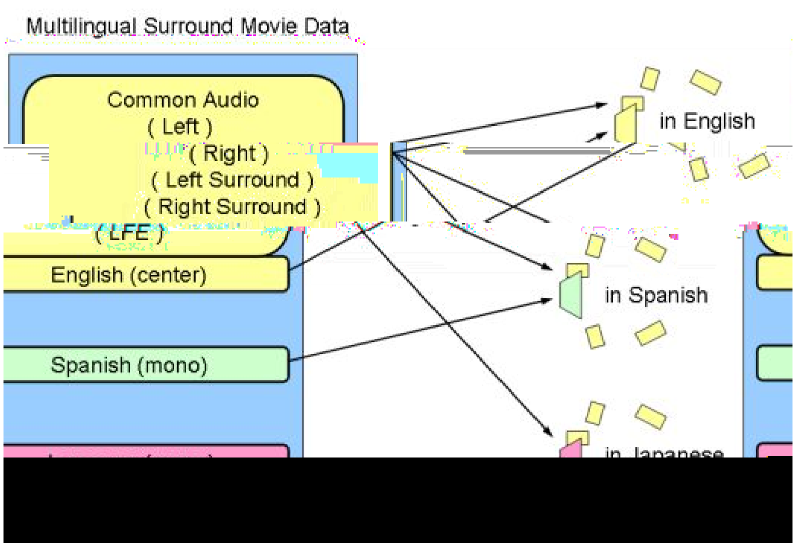
5.1

5.1

5.1

```
ns.audioTrack = 0;      // main 5.1 audio track
ns.voiceTrack = 5;      // mono audio track
```

5.1



センターチャネルの置き換え

再生

SubAudio … GFx::Video::Video … GFx::Video::Video … SubAudio …

```
ns.audioTrack = 0;       // main audio track
ns.subAudioTrack = 6;    // secondary audio track
```

… Sound.setVolume() …

```
var audio:Sound = new Sound(this);
audio.setVolume(80, 50);       // set volume for main and subaudio tracks
```

再生

トラブルシューティング

1. IO

2.

3.

3

1. スレッドの設定 – 720P 3 4 8

2. ビデオの解像度とビットレート – Scaleform

3. バッファーとフレームプール - ActionScript 2 6

```
a. NetStream.setNumberOfFramePools(numPools:Number)
b. NetStream.setBufferTime(bufferTime:Number)
```